WebClient Computing

Document Object Model (DOM)

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Announcements

- Project score is out
  - Note it is in the relative scale
  - Project score evaluation criteria in the course description
- Midterm + Project-1 score distribution is out
  - https://docs.google.com/spreadsheets/d/14j-GXJKQXey-hKk7eQxjy6o0T8UfeyebWvn-dRwfQiM/edit?usp=sharing

Defer Function Execution

- setTimeout(Function, elapsed_time)
  - After the elapsed_time in the second argument passes, the given function in the first argument is executed
  - Executes only once and exits
  - The unit of elapsed_time is in milli-seconds
  - Remember passing a function as an argument
    - Array.sort();
- setInterval(Function, period_time)
  - At every, period_time, perform Function
- clearInterval(id)
  - Cancel the periodic interval function

Defer Function Example

```
function setTimeoutFunc() {
  console.log("calling setTimeout function");
}

setTimeout(setTimeoutFunc, 2000);

var cid = setInterval(function() {console.log("calling setInterval")}, 3000);

setTimeout(function() {clearInterval(cid)}, 10000);
```

Window Object in Javascript

- An object to represent open window in a browser
  - A topmost object in a console or page
- You can see window object in the console tab
- Handful properties and methods
  - location, location.href, alert, screen, document, document.body, document.styleSheets (styleSheets[0].cssRules[0].style.cssText)
  - Full list - http://www.w3schools.com/jsref/obj_window.asp

Example

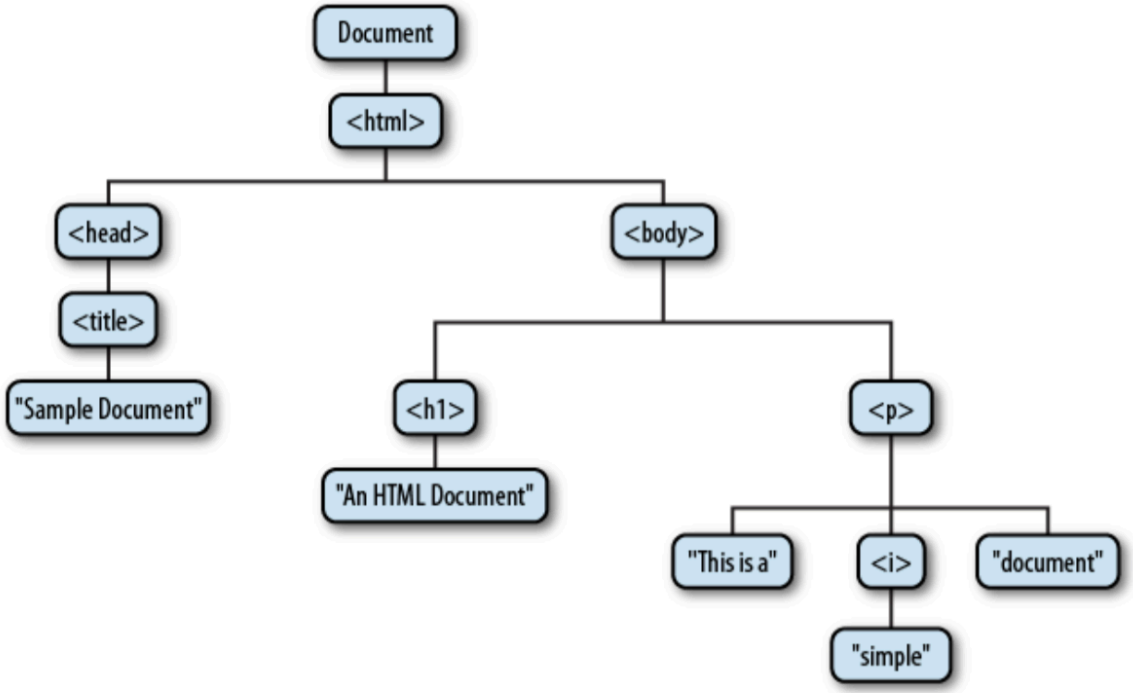
- window.js

```
> window.location
<· ▶ Location {hash: "", search: "", pathname:
   "/Users/kyungyonglee/Documents/Courses/2016-
   01/WebClient/Code/Javascript/window.html", port: "", hostname:
   ""…}
> window.location.href
<· "file:///Users/kyungyonglee/Documents/Courses/2016-
   01/WebClient/Code/Javascript/window.html"
> window.alert("popup")
<· undefined
> document.body
<· ▶<body>…</body>
> document.styleSheets[0].cssRules[0].style.cssText
<· "color: orange; text-align: center;"
> window.screen
<· ▶ Screen {availWidth: 1280, availHeight: 777, width: 1280, height:
   800, colorDepth: 24…}
```

DOM (Document Object Model)

- Represent HTML file as an object so that Javascript can easily access
- Programming interface for HTML
- Represented in a tree structure where the <html> tag is the root

```
<html>
  <head>
    <title>Sample Document</title>
  </head>
  <body>
    <h1>An HTML Document</h1>
    <p>This is a <i>simple</i> document.
</html>
```



Applications of DOM

- Javascript can change HTML by accessing DOM
  - Elements
  - Attributes
  - CSS style
- Javascript can add or remove existing elements and attributes
- Javascript can react to events happening in a page (ex. Button click)

https://www.javacodegeeks.com/2016/01/7-ways-es2015-can-improve-your-javascript-programming.html
https://en.wikipedia.org/wiki/HTML

Some Handful Document Properties

- document.lastModified
  - Date and time the document was last modified
- document.title
  - Returns the title of a document
- document.URL
  - Return the full URL of a given document
- element.style.visibility
  - Sets an element visibility - hidden or visible
- window.open, document.open(), document.write(), document.close()
  - Open a new window and add a message

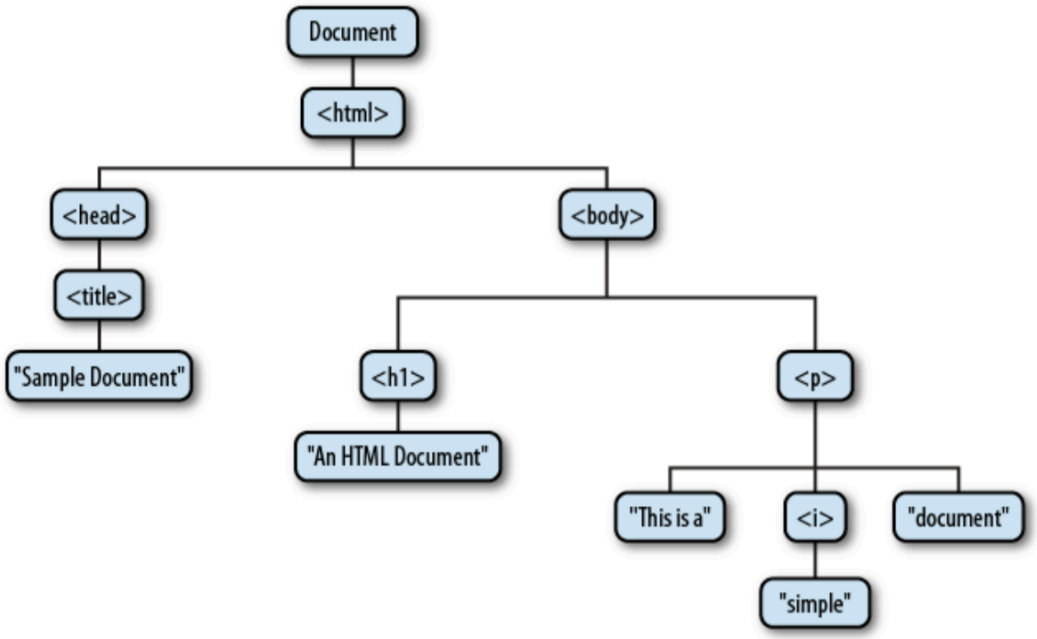
Examples

- window.html

```
> document.lastModified
<· "05/12/2016 14:29:48"
> document.title
<· "Heading test"
> document.URL
<· "file:///Users/kyungyonglee/Documents/Courses/2016-01/WebClient/Code/Javascript/window.html"
> document.getElementsByTagName("h1")[0].style.visibility='hidden';
<· "hidden"
> document.getElementsByTagName("h1")[0].style.visibility='visible';
<· "visible"
```

Element Node in a Tree

- Represents an HTML element
- Parent node
  - A node directly above a node
- Child nodes
  - Nodes one level directly below
- Sibling nodes
  - Nodes at the same level (same parent node)
- Descendant nodes
  - A set of nodes any number of levels below another node
- Ancestor nodes
  - A set of nodes above a node in a tree



Node in DOM

- Each entry in a tree is a Node
- Element node
  - Representation of element in a HTML <body>, <p>, …
- Attribute node
  - Attribute in an element <a href="">, <img src="">, ...
- Text node
  - Text value within an element
- Example
  - <a href="http://www.kookmin.ac.kr">KMU</a>

Element Attribute Text

Accessing Element

- document.getElementById(“ID_NAME”)
  - Returns an element with a given ID_NAME
  - Duplicate ID is not allowed in the HTML specification
- dom_element_by_id.html

```
> var t = document.getElementById("first-p");
<· undefined
> t
<· ▼<p id="first-p">
      "The "
      <strong>first</strong>
      " paragraph"
    </p>
> t.innerHTML
<· "The <strong>first</strong> paragraph"
> t.innerText
<· "The first paragraph"
```

```
<!DOCTYPE html>
<html>
<head>
  <title>DOM test</title>
</head>
<body>
  <p id = "first-p">The <strong>first</strong> paragraph</p>
  <p id = "second-p">The second paragraph</p>
  <p id = "first-p">Duplicated ID paragraph</p>
</body>
</html>
```

Changing Values of an Element

- Assign a value to an element
- Change style of an element
  - Using element.style property

```
> t.innerText = "Changing content";
<· "Changing content"
> t.style.fontSize = "3em";
<· "3em"
> t.style.color="red";
<· "red"
```

Changing content

The second paragraph

Duplicated ID paragraph

Access Elements By Tag

- document.getElementsByClassName
  - Get a list of elements based on the class name
- document.getElementsByName
  - Get a list of elements based on the name
  - Name is generally used with input tag
- document.getElementsByTagName
  - Get a list of elements with the input tag name

Accessing Elements Examples

```
> t=document.getElementsByClassName("class-1");
<· [  <p class="class-1">Class 1 paragraph</p>,
    <div class="class-1">Class 1 div</div>]
> t.length
<· 2
> t[0]
<·   <p class="class-1">Class 1 paragraph</p>
> t[1]
<·   <div class="class-1">Class 1 div</div>
```

```
> t=document.getElementsByTagName("p");
<· [  <p class="class-1">Class 1 paragraph</p>,
    <p class="class-2">Class 2 paragraph</p>]
```

```
> t=document.getElementsByName("text-input");
<· [  <input type="text" name="text-input">,
    <input type="text" name="text-input">]
> t[0]
<·   <input type="text" name="text-input">
> t[1].value = "Kyungyong";
<· "Kyungyong"
> t[1].value;
<· "Lee"
```

```
<!DOCTYPE html>
<html>
<head>
  <title>DOM test</title>
</head>
<body>
  <p class = "class-1">Class 1 paragraph</p>
  <p class = "class-2">Class 2 paragraph</p>
  <div class = "class-1">Class 1 div</div>
  ID: <input type="text" name="text-input">
  Name: <input type="text" name="text-input">
</body>
</html>
```

dom_elements_by.html

References

- Javascript, the Definitive Guide
- w3schools.com
- tryhelloworld.com